

VARUNA (SIH26143) — Master Viva & Technical Q&A Guide

30 Comprehensive Questions & Answers | System Architecture, Algorithms, SAR Preprocessing, Geodesy & Security

Section A: Project Overview, Scope & Life Cycle

Q1: What is VARUNA and what problem statement (SIH26143) does it solve?

VARUNA (**V**essel **A**tttribution through **R**emote-sensing & **U**nified **N**avigational **A**alytics) is an environmental maritime investigation platform built for Smart India Hackathon 2026 (Problem Statement **SIH26143**). When an illegal oil spill happens at sea, VARUNA detects it from Sentinel-1 SAR satellite imagery, uses backward ocean drift physics to find where the leak started, matches that area against real historical AIS ship GPS data, and ranks the suspect vessels in a printable, audit-proof evidence report.

Q2: What is the non-negotiable Real-Data Policy in VARUNA?

VARUNA enforces a strict **Zero Mock Data Policy**. Zero fake, synthetic, or placeholder data is allowed anywhere in the product, demos, screenshots, metrics, or model training. Every single data object (satellite scenes, oil detections, vessel tracks, origin estimates, candidates) carries a mandatory **provenance record** (sourceType, retrievedAt, externalId, checksum). Objects missing provenance are automatically stripped by API middleware guards.

Q3: What Software Development Life Cycle (SDLC) model was used to build VARUNA?

VARUNA was developed using a **Phase-Gated Iterative & Incremental Agile MVP Model**. The project was built across 14 sequential phases (Phase 0 to Phase 13). Each phase defined strict exit criteria, automated unit/integration tests, and real-data policy compliance before landing on main.

Q4: What is the complete technology stack of VARUNA?

- **Frontend:** React 18, Vite, TypeScript, MapLibre GL JS, deck.gl (WebGL visualization), React Three Fiber / Drei (3D graphics), TanStack Query, Vanilla CSS design tokens.
- **Backend API:** Node.js 20+, Express 5, TypeScript, Argon2id, JWT access cookies, opaque SHA-256 refresh tokens, Zod boundary validation, GeographicLib geodesy, Turf.js.
- **AI/ML Microservice:** Python 3.11, FastAPI, PyTorch, SciPy, scikit-image, rasterio, GDAL, pyproj, GeoPandas, Shapely.
- **Queues & Workers:** BullMQ on Redis 7, Puppeteer (headless Chrome PDF rendering).
- **Databases & Storage:** MongoDB 8 (Time-series collections for AIS, 2dsphere spatial indexes), Redis 7, MinIO (S3 object storage), TiTiler (dynamic Cloud-Optimized GeoTIFF raster server).

Section B: Satellite Imagery & Image Preprocessing

Q5: What satellite imagery is used and why Synthetic Aperture Radar (SAR)?

VARUNA uses **Sentinel-1 C-band Synthetic Aperture Radar (SAR)** imagery. Unlike optical satellite cameras, SAR radar penetrates cloud cover and fog, working day and night. Oil smooths out capillary waves on the sea surface, causing radar signals (σ^0) to bounce away from the satellite, making oil slicks appear as dark black spots on SAR imagery.

Q Q6: How does raw SAR satellite imagery get preprocessed before detection?

Preprocessing involves 5 key steps:

1. **Radiometric Terrain Correction (RTC):** Calibrates raw radar backscatter to linear σ^0 values.
2. **Decibel Conversion (to_db):** Converts linear σ^0 to decibels (dB) via $10 * \log_{10}(\sigma^0)$, setting invalid values ($\leq 10^{-6}$) to NaN.
3. **Land & Bright Target Masking:** Masks pixels > -8.0 dB (land and bright metal ships) using morphological closing and dilation.
4. **Scale-Invariant Background Estimation:** Downsamples by $32\times$, applies a 9×9 median filter, and scales back up to calculate the local sea background efficiently.
5. **Morphological Speckle Cleanup:** Applies morphological opening (disk 2), closing (disk 4), small object removal (< 0.05 km²), and interior hole filling.

Section C: Detection Algorithm & Review Workflow

Q Q7: What algorithm is used for oil slick detection (darkspot-v1)?

Detection uses **Classical Adaptive Local Thresholding (darkspot-v1)**. Pixels darker than their local sea background by ≥ 3.0 dB are segmented. Connected components are labeled, and shape/context features (area, perimeter, major/minor axes, elongation, convexity, contrast) are extracted to compute confidence.

Q Q8: Why was Adaptive Local Thresholding chosen over a global threshold?

A global threshold fails on SAR imagery because mean backscatter varies across a swath with incidence angle and local wind gradients. Adaptive local thresholding compares each pixel against a large local median background, making the segmentation scale-invariant across the entire scene.

Q Q9: How is the Look-Alike Risk score (R_look_alike) calculated?

It evaluates how much a dark spot resembles non-oil look-alikes (0 = Oil-like, 1 = Look-alike):

```
R_look_alike = clip(0.35 * R_shape + 0.25 * R_convexity + 0.25 * R_contrast + 0.15 * R_size, 0, 1)
```

Real slicks are drawn out by wind/current (high elongation), have ragged boundaries (low convexity), and high contrast.

Q Q10: What is the Wind Suitability Gate (W_suit) and why is it important?

Wind speed (u_{10}) directly affects SAR slick visibility:

- $u_{10} < 2$ m/s or > 14 m/s: $W_{suit} = 0.05$ (Glassy sea or slick re-roughened by high waves).
- 4 m/s $\leq u_{10} \leq 9$ m/s: $W_{suit} = 1.00$ (Optimal wind window).
- **Unknown wind:** $W_{suit} = 0.50$.

This term prevents low-wind calm sea zones from receiving high overall confidence.

Q Q11: How is the 4-term Detection Confidence score computed?

$$\text{Confidence} = 0.40 * \text{Separation} + 0.35 * \text{Shape} + 0.15 * \text{Wind} + 0.10 * \text{Size}$$

Where Separation = clip(contrast_dB / 10, 0, 1) and Shape = 1 - R_look_alike. Each term is reported separately so analysts see why confidence is high or low.

Q Q12: How does the Human-in-the-Loop review system work for detections?

An **UNREVIEWED** slick renders hatched. Analysts take review actions (CONFIRM, EDIT, REJECT):

- **EDIT:** Requires a corrected polygon and recomputes geodesic area.
- **REJECT:** Requires a note and a physical category (LOOK_ALIKE vs OPERATIONAL).
- Original model output (version 0) is immutable and preserved inside reviewHistory. Categorized look-alike rejections automatically generate labeled training datasets for future ML retrains.

Section D: Deep Learning Experiments & Empirical Metrics**Q Q13: Why was the deep learning U-Net model evaluated and rejected for production?**

A small U-Net (~0.5M parameters) trained on 315 scenes was evaluated on the 66 held-out test scenes. While it achieved good IoU on clear slicks, it exhibited a **higher false-positive rate on look-alikes (75%)** than the classical detector (68%). Following VARUNA's empirical quality rules, darkspot-v1 was retained for production.

Q Q14: What are the measured accuracy metrics of darkspot-v1 on the test set?

Evaluated on the Trujillo-Acatitla Part III held-out test split (66 scenes):

- **Oil Detection Rate:** 100% (22/22 oil scenes detected; 0 missed).
- **Mean Oil IoU:** 0.56 (Median: 0.63).
- **False-Positive Rate on Look-alikes:** 68% (Fired on 15 of 22 look-alikes).
- **False-Positive Rate on Clean Sea:** 18% (Fired on 4 of 22 clean sea scenes).

Q Q15: What datasets were used for training and testing VARUNA?

- **Segmentation Dataset:** Trujillo-Acatitla et al. (2024) Sentinel-1 Oil Spill Dataset Part III (CC-BY-4.0) — 450 dual-pol 2048×2048 decibel scenes.
- **Geographic Split:** 315 training scenes, 69 validation scenes, 66 held-out test scenes grouped by ocean grid cells.
- **AIS Data:** Real historical AIS data from NOAA Marine Cadastre (1-minute intervals) and Danish Maritime Authority (DMA).
- **Demo Incident:** Guam Apra Harbour (2025-09-21) with 9,711 real historical AIS fixes.

Section E: Geodesy & Reverse Ocean Drift Physics

Q Q16: Why is Karney's Ellipsoidal Geodesy used instead of Haversine?

Haversine assumes a spherical Earth and introduces up to ~0.5% distance/area error. Karney's algorithm calculates geodesics on the true WGS84 ellipsoid. Both JS (geographiclib-geodesic) and Python (pyproj.Geod) implement Karney's math and pass a shared known-answer test suite (geo-known-answers.json) within < 0.1% agreement.

Q Q17: How does Reverse Lagrangian Particle Drift Integration work?

To find where oil originated, 5,000 virtual particles are uniformly rejection-sampled across the slick footprint and integrated **backwards in time ($\Delta t = -15$ min)**:

$$dx/dt = u_{\text{current}} + \alpha * R(\theta) * u_{\text{wind}} + \text{RandomWalk}(K_h)$$

Where wind factor $\alpha \sim U(0.02, 0.04)$, Ekman deflection $\theta \sim U(0^\circ, 20^\circ)$, and eddy diffusivity $K_h = 10 \text{ m}^2/\text{s}$.

Q Q18: What is the difference between OK and DEGRADED Origin estimates?

- **OK:** Driven by real ocean current (CMEMS/HYCOM) and wind (ERA5/GFS) forcing data.
- **DEGRADED:** Triggered when forcing data is missing for the incident date. Uses a footprint proximity buffer, expands the AIS search envelope from 15 km to 40 km, marks feature F8 MISSING, and **hard-caps all candidate vessel ranks at MODERATE**.

Q Q19: How are Release Zones and Time Windows generated from particles?

Particles are processed using SciPy Gaussian Kernel Density Estimation (gaussian_kde) to produce support50 (50% probability) and support90 (90% probability) spatial contours. The release time window calculates 4 timestamps (earliest, latest, mostLikelyStart, mostLikelyEnd) and is constrained by prior clear satellite scenes.

Section F: AIS Ingestion & Vessel Attribution Model

Q Q20: How is historical AIS data ingested and cleaned in MongoDB?

AIS positions land in the `ais_positions` **time-series collection** (time field `t`, meta field `meta`, granularity seconds). Ingestion transforms clean sentinel values (SOG 102.3 kn $\rightarrow$ null, COG 360.0 $\rightarrow$ null, heading 511 $\rightarrow$ null), validates 9-digit MMSIs against ITU country MIDs, deduplicates records, and attaches vertex timestamps (`times[]`).

Q Q21: What is AIS Dark Period detection (F5) and why is it important?

AIS transmission dropouts over the release window are classic deliberate discharge signatures. Overlapping gaps are detected and measured: `f5 = min(1, gap_minutes / 120)`. Gaps that overlap the window feed feature F5, while non-overlapping gaps score a measured 0.

Q Q22: What are the 12 features in the Candidate Vessel Attribution model?

Features sum to 1.00 weight: F1 Spatial Proximity (0.18), F2 Temporal Alignment (0.16), F3 Track Intersection (0.13), F4 Heading Alignment (0.10), F5 AIS Dark Period (0.10), F6 Speed Consistency (0.08), F7 Vessel Type Prior (0.07), F8 Origin Density (0.07), F9 Draught Change (0.05), F10 Slick Axis Continuity (0.03), F11 Manoeuvre Anomaly (0.02), F12 Prior Incident History (0.01).

Q Q23: How does score renormalization work for candidate vessels?

$$\text{Score} = 100 * (\sum \text{Measured_Contributions}) / (\sum \text{Measured_Weights})$$

Renormalizing over measured weights ensures a vessel missing attributes in public AIS feeds (e.g., vessel type or draught) is not penalised against vessels with complete records.

Q Q24: What is the Insufficient Evidence Gate?

If fewer than **6 of the 12 features** can be measured due to data gaps, the candidate's classification tier is forced to **INSUFFICIENT_EVIDENCE**, regardless of numeric score. If the top candidate is **INSUFFICIENT_EVIDENCE**, the ranking leaderboard is replaced with a warning banner.

Q Q25: How are Bootstrap Confidence Intervals and Paired Resampling computed?

Monte Carlo bootstrap resampling (300 iterations) resamples the drift particle ensemble and interpolates un-fixed trajectory positions to compute 95% CIs (Score $\pm$ margin). Paired resampling draws the shared origin zone once per iteration to measure rank stability probability (e.g. Candidate #1 ranks first in 94.2% of draws).

Section G: Performance, Complexity, Security & Gaps**Q Q26: What are the Time and Space Complexities of VARUNA's core modules?**

- **SAR Segmentation:** Time $O(N)$, Space $O(N)$ where N = total pixels (~1 to 2 seconds).
- **Drift Integration:** Time $O(P * T)$ (~100 ms for $P=5000$ particles, $T=96$ steps), Space $O(P * T)$.
- **KDE Density Surface:** Time $O(P * G)$, Space $O(G)$ where G = grid cells.
- **Candidate Vessel Correlation:** Time $O(M * V * K)$ where $M=300$ resamples, V =vessels, K =fixes.

Q Q27: What key gaps in existing commercial systems does VARUNA solve?

Existing systems (CleanSeaNet, EMSA) provide slick detection but lack automated probabilistic vessel correlation, black-box scores without error margins, and report empty lists when AIS is missing. VARUNA provides probabilistic KDE release surfaces, 95% CIs, paired rank stability, explicit **NO_AIS_COVERAGE** warnings, and version-controlled audit histories.

Q Q28: How does VARUNA prevent dataset overfitting and underfitting?

Overfitting is controlled via geographic grid splitting (zero spatial overlap between train and test cells), parameter caps (~0.5M params for U-Net), and rotational data augmentation. Underfitting is prevented by combining Binary Cross-Entropy with Soft Dice Loss to prevent trivial 'predict all sea' solutions.

Q Q29: How does Authentication and Role-Based Access Control (RBAC) work?

Auth uses Argon2id password hashing, 15-minute access JWTs in httpOnly cookies, and 7-day opaque SHA-256 refresh tokens rotated on use (replaying spent tokens revokes the entire session family). RBAC ranks 4 roles: viewer (0) → analyst (1) → lead (2) → admin (3). Invisible resources return 404 Not Found rather than 403 Forbidden to prevent resource enumeration.

Q Q30: How does the Provider Federation Chain work for catalogue search?

Catalogue queries run across satellite providers (Copernicus CDSE, MS Planetary Computer, NASA ASF) in parallel with Redis quota tracking, exponential backoff, and 5-failure circuit breakers. Deduplication prioritizes preprocessed Radiometric Terrain Corrected (RTC) scenes. A provider failure advances the chain, but zero results do not—an empty answer is preserved as honest coverage.